

March 2026 results

Broken LCD (SLM Off)

For the initial TIE reconstruction, the LCD was used in an unmodulated state, with no applied Arduino control signal, effectively representing the SLM in an “off” condition. In this case, the LCD surface itself including inherent defects and structural features served as the source of intensity variation. Shown in figures 2-7.

The reconstruction reveals several consistent features of the physical LCD. In particular, a prominent defect located near the bottom of the screen, likely a scratch or dent, is clearly reproduced in the 3D surface profile. Additionally, diagonal line structures observed on the LCD surface are also visible in the reconstructed image, particularly in the upper region, indicating that the TIE method is capable of capturing real surface characteristics under these conditions. However, the overall reconstruction exhibits significant irregularities, appearing lumpy and wave-like rather than smooth. This behavior is consistent with earlier trials in which similar wave-like artifacts were observed. These distortions are likely due to noise in the system, including environmental factors such as air fluctuations, particle interactions, and instability in illumination when the SLM is not actively controlled. To mitigate these effects, noise reduction techniques were implemented in the MATLAB reconstruction code; however, residual artifacts remain, suggesting that noise continues to influence the recovered phase and height profiles. This

suggests that while TIE can recover true surface features, its accuracy is strongly affected by noise when intensity modulation is not well controlled.

Trial 1

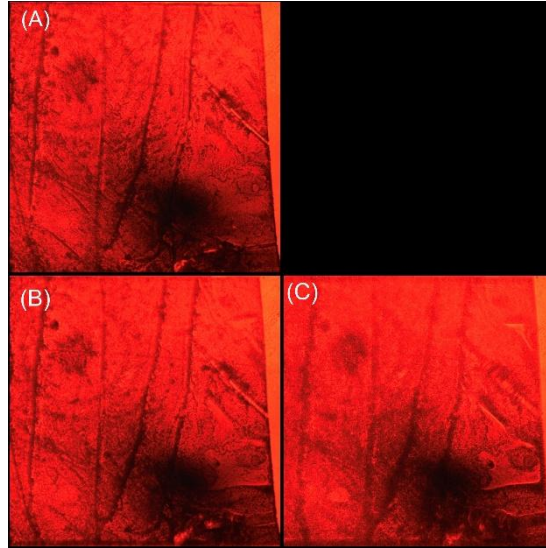


Figure 2 shows an Lcd scrap from peeling. (A) Primary image (B) 5mm ahead of I_0 (c) 5mm behind I_0

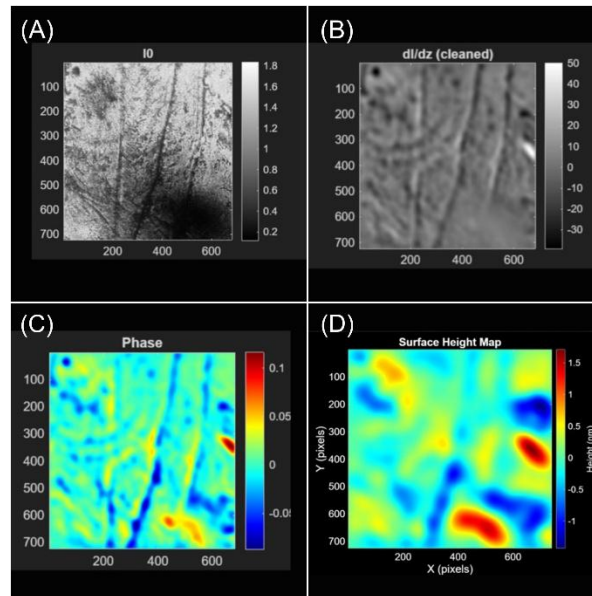


Figure 3 shows (A) In-focus intensity image I_0 , showing inherent surface texture and illumination variations. (B) Axial intensity derivative $\partial I / \partial z$, highlighting edges and structural features. (C) Recovered phase distribution, representing optical path variations across the surface. (D) Corresponding surface height map, where localized peaks and valleys reflect surface irregularities of the LCD.

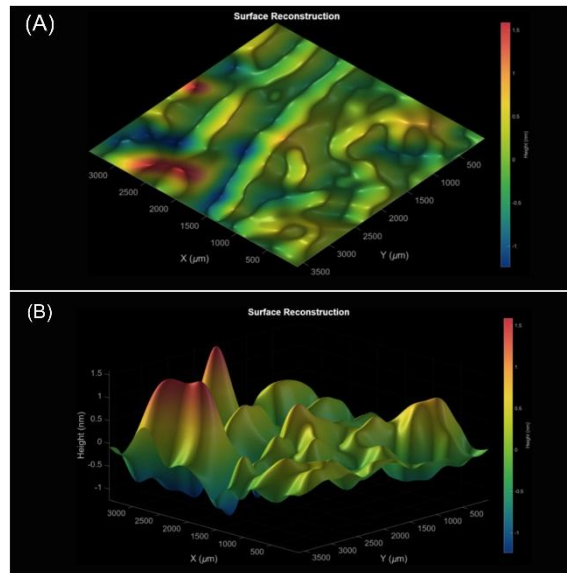


Figure 4 shows the 3d reconstruction of the lcd slm turned off and shows characteristics of the damage. (A) top view (B) side view

Trial 2

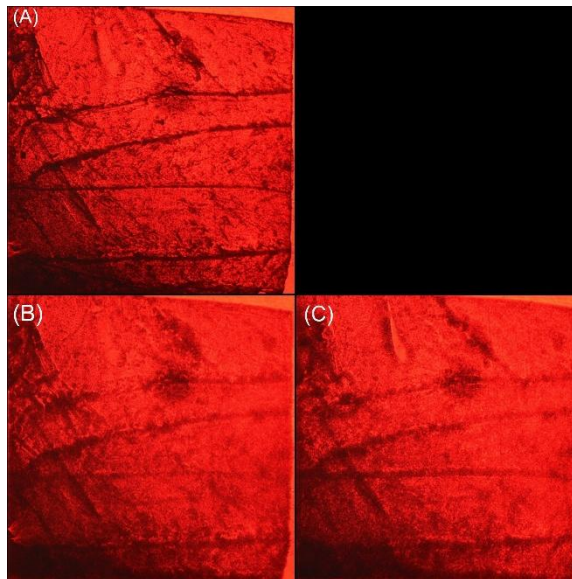


Figure 5 shows an Lcd scrap from peeling. (A) Primary image (B) 5mm ahead of I0 (c) 5mm behind I0

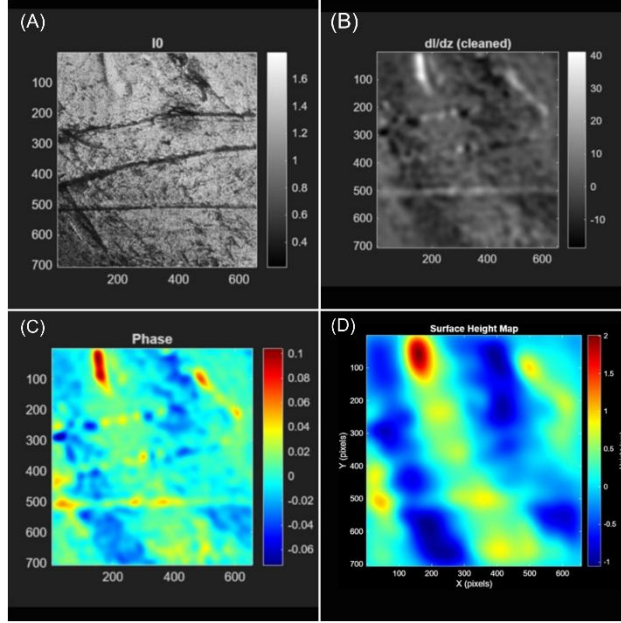


Figure 6 shows Figure 3 shows (A) In-focus intensity image I_0 , showing inherent surface texture and illumination variations. (B) Axial intensity derivative $\partial I / \partial z$, highlighting edges and structural features. (C) Recovered phase distribution, representing optical path variations across the surface. (D) Corresponding surface height map, where localized peaks and valleys reflect surface irregularities of the LCD.

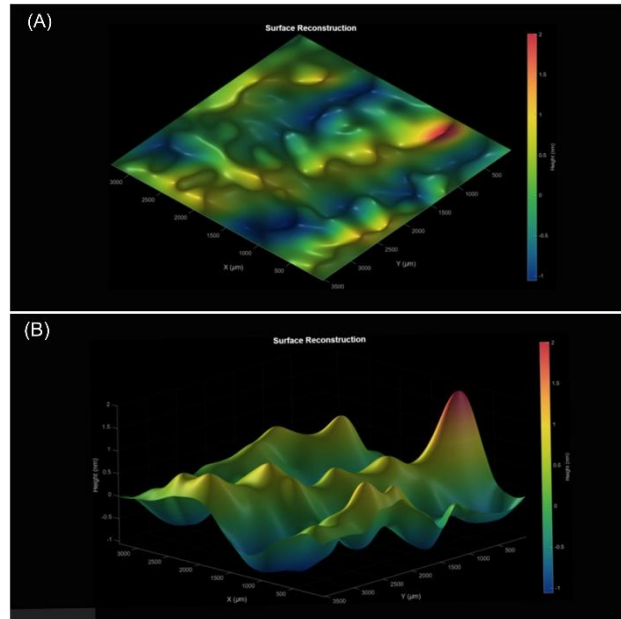


Figure 7 shows the 3d reconstruction of the lcd slm turned off and shows characteristics of the damage. (A) top view (B) side view